

Disproof of conjecture 00000001556

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2026-09-14

Verdict: FALSE

We disprove conjecture 00000001556 as stated. The refutation is unconditional and elementary: the explicit map

$$c_5(x, y) = (x + 2y) \bmod 5$$

is a proper 5-colouring of the lattice distance graph for *both* standard readings of the distance set $D = \{1, 2, 4\}$ simultaneously. Hence the chromatic number is at most 5, and the claimed value 7 is false. In fact the exact values are 5 (squared-norm reading) and 3 (Euclidean-norm reading).

1 The conjecture, quoted verbatim

From `conjectures/00000001556.md`:

English. Conjecture: The chromatic number of the planar distance graph $G(\mathbb{Z}^2, D)$ with $D = \{1, 2, 4\}$ is 7 (the exact chromatic number of a fixed small distance set).

The conjecture is filed in both English and Chinese. The Chinese version is reproduced verbatim (in Unicode) in `README.md`; the two versions are identical in content. The PDF font does not embed CJK glyphs, so the Chinese is kept in the markdown copy rather than duplicated here.

2 Definitions and the definitional ambiguity

Definition 1 (Distance graph). $G(\mathbb{Z}^2, D)$ has vertex set $\mathbb{Z}^2$, and two lattice points u, v are adjacent iff the difference $u - v$ realises a “distance” in D .

The conjecture is written with three scalars $D = \{1, 2, 4\}$, so D is a set of distances, not of integer vectors. There are two standard and equally natural readings:

- **Squared-norm reading.** Lattice-colouring work often states a distance set as squared norms, because squared distances are integral on the lattice. Here (dx, dy) is a displacement iff

$$dx^2 + dy^2 \in \{1, 2, 4\}.$$

This gives the 12 vectors $(\pm 1, 0), (0, \pm 1), (\pm 1, \pm 1), (0, \pm 2), (\pm 2, 0)$.

- **Euclidean-norm reading.** The literal reading: (dx, dy) is a displacement iff

$$\sqrt{dx^2 + dy^2} \in \{1, 2, 4\},$$

i.e. $dx^2 + dy^2 \in \{1, 4, 16\}$. This gives the 12 axis vectors

$$(\pm 1, 0), (0, \pm 1), (\pm 2, 0), (0, \pm 2), (\pm 4, 0), (0, \pm 4).$$

The union of the two displacement sets has 16 vectors:

$$\begin{aligned} D_{\text{sq}} &= \{(\pm 1, 0), (0, \pm 1), (\pm 1, \pm 1), (\pm 2, 0), (0, \pm 2)\}, & |D_{\text{sq}}| &= 12, \\ D_{\text{euc}} &= \{(\pm 1, 0), (0, \pm 1), (\pm 2, 0), (0, \pm 2), (\pm 4, 0), (0, \pm 4)\}, & |D_{\text{euc}}| &= 12, \\ D_{\text{union}} &= D_{\text{sq}} \cup D_{\text{euc}}, & |D_{\text{union}}| &= 16. \end{aligned}$$

The four extra vectors are $(\pm 4, 0), (0, \pm 4)$.

Remark 2 (“Planar”). Here “planar distance graph” names a graph drawn on the integer lattice, not a planar graph in the graph-theoretic sense; $G(\mathbb{Z}^2, D)$ is far from planar.

3 The witness colouring

Theorem 3. Define $c_5: \mathbb{Z}^2 \rightarrow \{0, 1, 2, 3, 4\}$ by $c_5(x, y) = (x + 2y) \bmod 5$. Then c_5 is a proper colouring of the union graph $G(\mathbb{Z}^2, D_{\text{union}})$, and therefore of $G(\mathbb{Z}^2, D_{\text{sq}})$ and $G(\mathbb{Z}^2, D_{\text{euc}})$ as well. Consequently

$$\chi(G(\mathbb{Z}^2, D)) \leq 5 < 7$$

under both standard readings, and the conjectured value 7 is false.

Proof. Suppose $u = (x, y)$ and $u + d = (x + dx, y + dy)$ are adjacent for some $d = (dx, dy) \in D_{\text{union}}$. Then

$$c_5(u + d) - c_5(u) \equiv (x + dx) + 2(y + dy) - (x + 2y) \equiv dx + 2dy \pmod{5}.$$

The displacements of the union and the corresponding residues $dx + 2dy \bmod 5$ are listed in the table below; none is 0 (mod 5). Hence $c_5(u + d) \neq c_5(u)$ for every adjacent pair, so c_5 is proper. Since it uses 5 colours, $\chi \leq 5$. Every edge of either reading is an edge of the union graph, so the same colouring is proper for each reading. $\square$

displacement $d = (dx, dy)$	reading(s)	$dx + 2dy$	mod 5	$dx^2 + dy^2$	Euclidean length
(1, 0)	both	1	1	1	1
(-1, 0)	both	-1	4	1	1
(0, 1)	both	2	2	1	1
(0, -1)	both	-2	3	1	1
(1, 1)	squared	3	3	2	$\sqrt{2}$
(1, -1)	squared	-1	4	2	$\sqrt{2}$
(-1, 1)	squared	1	1	2	$\sqrt{2}$
(-1, -1)	squared	-3	2	2	$\sqrt{2}$
(2, 0)	both	2	2	4	2
(-2, 0)	both	-2	3	4	2
(0, 2)	both	4	4	4	2
(0, -2)	both	-4	1	4	2
(4, 0)	euclid	4	4	16	4
(-4, 0)	euclid	-4	1	16	4
(0, 4)	euclid	8	3	16	4
(0, -4)	euclid	-8	2	16	4

All sixteen residues are nonzero modulo 5, which proves Theorem 3.

Remark 4 (Periodicity and the de Bruijn–Erdős argument). The colouring c_5 is defined by a formula on all of $\mathbb{Z}^2$, so it needs no compactness argument to be global. Nevertheless, the general principle underlying lattice colourings is worth recording: by the de Bruijn–Erdős compactness theorem, the chromatic number of a (locally finite) graph is the supremum of the chromatic numbers of its finite subgraphs. For $G(\mathbb{Z}^2, D)$ this says $\chi = \sup_{F \subset \mathbb{Z}^2 \text{ finite}} \chi(F)$, so a finite clique already certifies a lower bound and a periodic colouring certifies a global upper bound. The two bounds below exploit exactly this.

4 Exact values, with cliques as lower bounds

Squared-norm reading: $\chi = 5$

Lower bound. The five points

$$C_5 = \{(0, 0), (1, 0), (-1, 0), (0, 1), (0, -1)\}$$

form a clique: the ten pairwise differences are $(\pm 1, 0), (0, \pm 1), (0, \pm 2), (\pm 2, 0), (\pm 1, \pm 1), (\pm 1, \mp 1)$, all of squared norm 1, 2 or 4. Hence $\chi \geq 5$. (This is the centre of the lattice together with its four nearest neighbours.)

Upper bound. Theorem 3 gives $\chi \leq 5$ via c_5 . Therefore $\chi = 5$ exactly.

Euclidean-norm reading: $\chi = 3$

Lower bound. The three points

$$C_3 = \{(0, 0), (1, 0), (2, 0)\}$$

form a clique: the pairwise differences are $(1, 0), (2, 0), (1, 0)$ of Euclidean length 1, 2, 1, all in $\{1, 2, 4\}$. Hence $\chi \geq 3$.

Upper bound. The map

$$c_3(x, y) = (x + y) \bmod 3$$

is proper: for a displacement $d = (dx, dy) \in D_{\text{euc}}$ we have $dx + dy \equiv \pm 1, \pm 2, \pm 4 \not\equiv 0 \pmod{3}$. Therefore $\chi \leq 3$, and $\chi = 3$ exactly.

Union graph: $\chi = 5$

The clique C_5 lies in the squared reading, hence in the union graph, giving $\chi \geq 5$; the colouring c_5 gives $\chi \leq 5$. So the union graph also has chromatic number exactly 5.

Corollary 5. *Under either standard reading of $D = \{1, 2, 4\}$, the conjectured value 7 is incorrect; the true values are 5 (squared) and 3 (Euclidean).*

5 Caveats: Manhattan and the continuous plane

A non-standard Manhattan reading gives $\chi \geq 9 > 7$

If instead D is read as a set of *Manhattan* (taxicab) distances $|dx| + |dy| \in \{1, 2, 4\}$, the displacement set is strictly larger than the union above (it adds $(\pm 1, \pm 3), (\pm 3, \pm 1), (\pm 2, \pm 2)$). The nine points

$$\{(0, 0), (1, \pm 1), (2, 0), (2, \pm 2), (3, \pm 1), (4, 0)\}$$

are pairwise at Manhattan distance 1, 2 or 4, so they form a 9-clique and $\chi \geq 9$. The colouring c_5 is *not* proper here, since the displacement $(3, 1)$ has $3 + 2 \cdot 1 = 5 \equiv 0 \pmod{5}$. Thus 7 is false under this reading too.

The continuous plane $\mathbb{R}^2$ is a different problem

The only reading that would rescue the value 7 is to reinterpret the vertex set $\mathbb{Z}^2$ as the continuous plane $\mathbb{R}^2$. There, the chromatic number of the plane with forbidden distance set $\{1, 2, 4\}$ is *not* determined by the elementary argument above; the Hadwiger–Nelson problem leaves the chromatic number of the plane in $\{5, 6, 7\}$, so 7 cannot be excluded in the continuum. This is not the conjecture as filed: the conjecture explicitly writes $G(\mathbb{Z}^2, D)$, the integer lattice, and for the lattice the refutation stands. A referee should read the present note as disproving the lattice statement, not the (open) continuum question.

6 Reproducibility

All claims above are checked by `reproduce.py` (Python 3 standard library only):

```
$ python3 reproduce.py
... (displacement sets for both readings, the 16-row residue table,
brute-force properness of  $c_5$  on  $[-200, 200]^2$  and of  $c_3$  on
 $[-100, 100]^2$ , the 5-clique and 3-clique, the Manhattan 9-clique)
PASS: all checks verified; conjecture 00000001556 is FALSE.
```

It prints PASS and exits 0 exactly when every check holds, and exits non-zero otherwise.

The LaTeX document is compiled with

```
tectonic --outdir build main.tex
```

producing `build/main.pdf`.

7 Formalisation

The refutation is formalised in the Lean 4 project under `lean4/` (core Lean only, `import Std`, no `Mathlib`, no `sorry`). The displacement list `D` is the 16-vector union of both readings, and the colourings are

```
def color (p : Int × Int) : Int := (p.1 + 2 * p.2) % 5
def color3 (p : Int × Int) : Int := (p.1 + p.2) % 3
```

The main statements are

```
theorem D_ok :
  D.all (fun e => decide ((e.1 + 2*e.2) % 5 ≠ 0)) = true
theorem color_shift_ne (x y : Int) (d : Int × Int)
  (hd : d ∈ D) :
  color (x + d.1, y + d.2) ≠ color (x, y)
theorem C5_clique :
  C5.all (fun p => C5.all (fun q => pairAdj p q)) = true
theorem Deuc_ok :
```

```

Deuc.all (fun e => decide ((e.1 + e.2) % 3 ≠ 0)) = true
theorem color3_shift_ne (x y : Int) (d : Int × Int)
  (hd : d ∈ Deuc) :
  color3 (x + d.1, y + d.2) ≠ color3 (x, y)
theorem C3_clique :
  C3.all (fun p => C3.all (fun q => pairAdj p q)) = true

```

where `pairAdj p q` is `p == q || D.contains (q.1 - p.1, q.2 - p.2)`. The residue checks and the clique certificates are pure `decide` computations; the properness theorems extract the nonzero residue from `D_ok` via `List.all_eq_true` and close with `omega`. See `lean4/README.md` for the statement table, the axiom audit, and the scope note.

Status against the submission rules

Rule 3 requires a LaTeX source, a PDF document, and a Lean 4 project.

- **LaTeX source** — `main.tex`, a standalone article using `geometry`, `amsmath`, `amssymb`, `amsthm`, `array`, `parskip`, `booktabs`; compiles with `tectonic main.tex`.
- **PDF** — at `build/main.pdf`, produced by `tectonic --outdir build main.tex`.
- **Lean 4 project** — `lean4/` with `lean-toolchain` pinned to `leanprover/lean4:v4.33.1`, `lakefile.toml` (library `Main`, no dependencies), `Main.lean`, `Check.lean` (`#print axioms`), and `lean4/README.md`. `lake build` exits 0 and the audit reports no `sorryAx`.